

Java Viva Summary Notes

Unit 1: Programming in Java

- Java Architecture: Source(.java) -> javac -> Bytecode(.class) -> JVM executes. JDK=development, JRE=runtime, JVM=executes bytecode.
- Buzzwords: Simple, OOP, Platform Independent, Secure, Robust, Multithreaded, Portable, Distributed, Dynamic.
- PATH locates Java commands; CLASSPATH locates classes/JARs.
- Arrays store same-type elements. Enhanced for loop: for(type x:array).
- Class=blueprint, Object=instance. Overloading=compile-time. Overriding=runtime.
- Access modifiers: public, protected, default, private. Interface defines contract. Inner classes: member, static nested, local, anonymous.
- final variable constant; final method cannot override; final class cannot inherit. static belongs to class.
- Packages organize classes.
- Exceptions: try, catch, finally, throw, throws. Custom exception extends Exception.
- Threads: states-New,Runnable,Running,Blocked/Waiting,Terminated. Create by Thread/Runnable. Synchronization prevents race conditions.
- Files: Byte streams(Input/OutputStream), Character streams(Reader/Writer), RandomAccessFile, Object streams for serialization.

Unit 2: Swing

- AWT is heavyweight; Swing lightweight and richer.
- Applet lifecycle: init(), start(), paint(), stop(), destroy().
- Layouts: Null(no manager), Flow, Border(N,S,E,W,C), Grid, GridBag(flexible), Group.
- Controls: JLabel, JTextField, JPasswordField, JTextArea, JScrollPane, JCheckBox, JRadioButton(ButtonGroup), JComboBox, JSlider.
- Menus: JMenuBar, JMenu, JMenuItem, PopupMenu, Toolbar, Tooltip, Mnemonics(Alt+key), Accelerators(shortcuts).
- Dialogs: JOptionPane, JDialog, JFileChooser, JColorChooser, JTable, JTree, JInternalFrame.

Unit 3: Event Handling

- Event generated by user action. Listeners: Action, Mouse, Key, Focus, Window, Item. Adapter classes provide empty implementations.

Unit 4: JDBC

- Architecture: Java App->JDBC API->DriverManager->Driver->DB.
- Driver types: Type1 JDBC-ODBC, Type2 Native, Type3 Network, Type4 Thin.
- Steps: Load driver, connect, Statement/PreparedStatement, execute, ResultSet, close.
- DDL: CREATE ALTER DROP. DML: INSERT UPDATE DELETE SELECT.

- PreparedStatement prevents SQL injection. ResultSets: forward, scrollable, updateable. Transactions: commit/rollback.

Unit 5: Networking

- TCP reliable connection-oriented. UDP faster connectionless.
- Socket enables communication. URL identifies resource. URLConnection communicates. JavaMail sends/receives email.

Unit 6: JavaFX

- Modern GUI replacing Swing. Layouts: FlowPane, BorderPane, HBox, VBox, GridPane. Controls: Label, TextField, Button, RadioButton, CheckBox, Hyperlink, Menu, Tooltip, FileChooser.

Unit 7: Servlets & JSP

- Servlet lifecycle: init(), service(), destroy(). GET retrieves, POST submits.
- Use HttpServletRequest/Response. Sessions store user data; Cookies stored in browser.
- JSP is view technology. Scripting: Declaration, Scriptlet, Expression. Implicit objects: request, response, session, application, out, config, page, pageContext, exception.
- Servlet vs JSP: logic vs presentation.

Unit 8: RMI & CORBA

- RMI allows Java remote method calls. Architecture: Client, Stub, Registry, Server.
- CORBA enables language-independent distributed objects using IDL. RMI Java-only; CORBA multilingual.